Universität Freiburg – Institut für Physik

Applied Theoretical Physics

Prof. Dr. Michael Moseler

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Due: 04.05.2026

Computational Physics: Materials Science - Exercise 1, SS 2026

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Tutorials: Mr. Andreas Doell

Problem 1. Projectile dynamics: analytical vs. numerical integration

In this exercise, we study the motion of a particle of mass m with initial velocity v_0 which is launched into the air at some angle θ in a gravitational field.

Use an AI of your choosing to implement such a Python simulation and compare numerical to analytical results. Try to code in a *modular* fashion. The goal of the tutorial is for you to use AI as a supportive tool, not as a replacement for learning. You are responsible to think critically and evaluate the results which you submit. Your report should demonstrate that you can explain your implementation and assess whether outputs are physically and numerically plausible.

Implementation

- For the numerical part implement two time-integration schemes:
  - Explicit Euler,
  - Velocity Verlet.

Analysis

Analyse the total energy of the system $E = T + U$. What happens when you switch from Euler to Velocity Verlet?

Additionally, analyse the following:

- The time of flight analytically and numerically.
- Repeat the simulation for different time steps Δt and discuss their impact on the simulation.
- What happens if you introduce air resistance to your model? How would you do that?

Play around with your parameters and observe how the results change. What happens if you are on Mars?

What to submit

Hand in a zip file on ILIAS containing the following files

- Python source code
- A short report in pdf-format with your main results

A good submission contains the following:

- A short description of the code and the goal of the exercise.

- Brief description how you managed to accomplish the task.
- A *selection* of plots which demonstrate that your code works.